

APP82661: Foundations of Directed Type Theory: Bridging Category Theory and Homotopy Type Theory

Application overview

Response deadline: 20 May 2026 (9 days left)

This is what your application will look like.

Foundations of Directed Type Theory: Bridging Category Theory and Homotopy Type Theory

1. Details

Summary

Type theory serves simultaneously as a logic, an alternative foundation for mathematics, and a programming language. It has become an increasingly powerful framework for the formal verification of both software systems and mathematical proofs. A major breakthrough in the last decade came with the development of Homotopy Type Theory (HoTT), which allows isomorphic structures to be treated as equal—a common practice in informal mathematics but previously challenging in formal systems. However, HoTT fundamentally relies on **symmetry**; it treats equivalences, not general functions, as primitive.

We propose a new and more general foundation: **Directed Type Theory (DTT)**. By taking functions (morphisms) rather than equivalences (isomorphisms) as the basic notion, DTT offers a directed perspective on types that aligns more closely with concepts from category theory and

higher-dimensional algebra. This promises a unifying framework that bridges HoTT and higher categories, potentially reshaping the theoretical underpinnings of both mathematics and computer science.

DTT opens the door to addressing several foundational questions:

- Parametricity and abstraction.

The univalence axiom reflects the abstractness of types, a key principle in both logic and computer science. However, as Reynolds observed, true abstraction also involves **parametricity**. We conjecture that a directed setting offers the right tools to reconcile univalence with parametricity, thereby deepening our understanding of abstraction in type theory.

- Core data type formers.

While HoTT provides many complex and expressive data types—such as multisets, real numbers, and type theory itself—it lacks a single type-former (analogous to the W-type for well-founded trees) that can uniformly express them. We believe DTT enables the formulation of such a unifying type-former, paving the way for a smaller and more principled core theory.

- Applications in quantum foundations.

Physicists exploring the mathematical foundations of quantum field theory and quantum gravity are increasingly turning to HoTT for formalisation. In this context, type theory components are reinterpreted in domain-specific ways, forming a framework for mechanised synthetic mathematics. We conjecture that DTT offers essential conceptual tools to handle directionality and non-reversibility inherent in these physical theories.

Our recent work has contributed to the emerging literature on DTT, but key questions remain open. This project will:

- * Formulate DTT in a way that allows coherent interaction with traditional (symmetric) type theories,
- * Develop the model-theoretic semantics of DTT, grounding it in category-theoretic and homotopical frameworks,
- * Establish core metatheoretic properties, such as decidability of type checking and normalisation,
- * Prototype an implementation demonstrating the practical feasibility of DTT as a foundation for computation and reasoning,
- * Explore concrete applications, including those outlined above, and

investigate connections with other frameworks, such as Higher Observational Type Theory (HOTT) and recent systems like Narya.

Methodologically, we will leverage existing proof assistants such as Agda, Idris, Lean or Rocq to formalise the metatheory, and we will build upon recent experimental platforms, notably Shulman’s Narya, to guide and validate the implementation of DTT.

This research aims to lay the foundational groundwork for a new generation of type theories, offering both deep theoretical insights and a platform for practical formal reasoning in mathematics, computer science, and physics.

Start date

October 2026

Duration

48 months

2. Core team

Name	Role	Organisation
Thorsten Altenkirch	Project lead	University of Nottingham
Nicolai Kraus	Project co-lead (UK)	University of Nottingham
Ulrik Buchholtz	Project co-lead (UK)	University of Nottingham

3. Thematic area alignment

Application question

Select the EPSRC thematic area your application most closely aligns to. This should be the theme where the majority of the novelty in the research lies.

In the text box, copy the letter corresponding to your selected theme:

- A. artificial intelligence (AI)
- B. digital security and resilience
- C. energy and decarbonisation
- D. engineering
- E. healthcare technologies
- F. information and communication technologies (ICT)
- G. manufacturing
- H. mathematical sciences
- I. physical sciences
- J. quantum technologies

Additional guidance:

This is for administrative purposes to help the initial application processing. We will check your choice and make a final decision on which theme will lead the peer review of your application.

We will not publish closing dates for individual rounds, instead we will batch applications internally. You should apply when your application is ready for submission and not be concerned about a closing date.

Applications will be batched for assessment and sent to the most appropriate responsive mode panel meeting when they reach this stage of the process.

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4. Vision and Approach

Application question

What are you hoping to achieve with and how will you deliver your proposed work?

Create a document that includes your responses to all criteria. The document should not be more than six sides of A4, single spaced in paper in 11-point Arial (or equivalent sans serif font) with margins of at least 2cm. You may include images, graphs, tables. You can have an additional one page for a diagrammatic workplan.

Upload your response as a single PDF document, see details in the file upload guidance. Enter the words 'attachment supplied' in the text box.

For the Vision, explain how your proposed work:

- is of excellent quality and importance within or beyond the field(s) or area(s)
- has the potential to advance current understanding, generates new knowledge, thinking or discovery within or beyond the field or area
- is timely given current trends, context and needs
- impacts world-leading research, society, the economy or the environment

Within the Vision section we also expect you to:

- identify the potential direct or indirect benefits and who the beneficiaries might be

For the Approach, explain how you have designed your work so that it:

- is effective and appropriate to achieve your objectives
- is feasible, and comprehensively identifies any risks to delivery and how they will be managed
- if applicable, uses a clear and transparent methodology
- if applicable, summarises the previous work and describes how this will be built upon and progressed
- will maximise translation of outputs into outcomes and impacts
- describes how your, and if applicable your team's, research environment (in terms of the place, and relevance to the project) will contribute to the success of the work

Within the Approach section we also expect you to:

- demonstrate access to the appropriate services, facilities, infrastructure, or equipment to deliver the proposed work
- provide a project plan including milestones and timelines in the form of a Gantt chart or similar (additional one-page A4)

See attached pdf, references are in section 6.

[4.Vision_and_Approach.pdf](#)

0.14MB

5. Applicant and team capability to deliver

Application question

Why are you the right individual or team to successfully deliver the proposed work?

Evidence of how you, and if relevant your team, have:

- the relevant experience (appropriate to career stage) to deliver the proposed work
- the right balance of skills and expertise to cover the proposed work
- the appropriate leadership and management skills to deliver the work and your approach to develop others
- contributed to developing a positive research environment and wider community

The word count for this section is 1,500 words, 1,000 words to be used for R4RI modules and, if necessary, a further 500 words for Additions.

Use the Résumé for Research and Innovation (R4RI) format to showcase the range of relevant skills you and, if relevant, your team (project and project co-leads, researchers, technicians, specialists, partners and so on) have and how this will help deliver the proposed work. You can include individuals' specific achievements but only choose past contributions that best evidence their ability to deliver this work.

Complete this section using the R4RI module headings listed. Use each heading once and include a response for the whole team, see the [UKRI](#)

[guidance on R4RI \(https://www.ukri.org/apply-for-funding/before-you-apply/resume-for-research-and-innovation-r4ri-guidance/#:~:text=UKRI%20committed%20to%20adopting%20a,UKRI%20throughout%202022%20and%202023.\)](https://www.ukri.org/apply-for-funding/before-you-apply/resume-for-research-and-innovation-r4ri-guidance/#:~:text=UKRI%20committed%20to%20adopting%20a,UKRI%20throughout%202022%20and%202023.). You should consider how to balance your answer, and emphasise where appropriate the key skills each team member brings:

- contributions to the generation of new ideas, tools, methodologies, or knowledge
- the development of others and maintenance of effective working relationships
- contributions to the wider research and innovation community
- contributions to broader research or innovation users and audiences and towards wider societal benefit

Additions

Provide any further details relevant to your application. This section is optional and can be up to 500 words. You should not use it to describe additional skills, experiences, or outputs, but you can use it to describe any factors that provide context for the rest of your R4RI (for example, details of career breaks if you wish to disclose them).

Complete this as a narrative. Do not format it like a CV.

Within this section you can also demonstrate elements of your responses in visual form if relevant. You should:

- use images sparingly and only to convey important information that cannot easily be put into words
- insert each new image onto a new line
- provide a descriptive legend for each image immediately underneath it (this counts towards your word limit)
- ensure that files are smaller than 8MB and in JPEG, JPG, JPE, JFI, JIF, JFIF, PNG, GIF, BMP or WEBP format

UK Research and Innovation (UKRI) has introduced new role types for funding opportunities being run on the new UKRI Funding Service.

For full details, see [Eligibility as an individual \(https://www.ukri.org/apply-for-funding/before-you-apply/check-if-you-are-eligible-for-research-and-innovation-funding/eligibility-as-an-individual/#contents-list\)](https://www.ukri.org/apply-for-funding/before-you-apply/check-if-you-are-eligible-for-research-and-innovation-funding/eligibility-as-an-individual/#contents-list).

The project leadership team consists of Thorsten Altenkirch (Professor of Computer Science), Ulrik Buchholtz (Assistant Professor of Computer Science), and Nicolai Kraus (Royal Society University Research Fellow and Professor of Theoretical Computer Science). In addition, our team comprises one full-time PDRA and nine project partners.

The proposal requires expertise in several different subfields and we, the leadership team, bring exactly this diverse blend of complementary skills to the project. Altenkirch has long-standing experience in the syntax and semantics of dependent type theories, Buchholtz is a leading expert in synthetic approaches to mathematics, and Kraus specialises on constructive higher structures. Our project partners are leading experts in their respective subfields and have a track record of successful collaborations with us.

Module 1 – Contributions to the generation of new ideas, tools, methodologies or knowledge

We and our project collaborators have a long history of successful collaborations which led to many joint publications. The Functional Programming Lab at the University of Nottingham, co-chaired by Altenkirch (PI/lead), is one of the internationally leading groups in the field of type theory. Altenkirch's main research interests are type theory, functional programming, categorical methods and quantum computing. He has published over 80 papers in these areas which are frequently cited (h-index 39). Similarly, Buchholtz and Kraus (co-leads) very frequently publish at leading venues in theoretical computer Science such as the flagship and CORE A*-rated conference LICS (logic in computer science), where, earlier this year, we received distinguished paper awards for two separate papers, providing a recent example of how our team's work is recognised at the highest levels in the wider field. For his contributions to homotopy type theory, Kraus further received the Ackermann Award 2016.

Specifically relevant for this project is Altenkirch's work with Neumann on directed CwFs [AN24]; Buchholtz's with Gratzner and Weinberger on Simplicial Type Theory including the distinguished LICS paper [GWB25]; and Kraus' paper on higher models of Type Theory [Kra21].

Module 2 – The development of others and maintenance of effective working relationships

We have ample experience leading research projects. Altenkirch served as the PI on on four EPSRC projects including *Homotopy Type Theory: Programming and Verification* as well as the US Airforce grant *Certified Programming with Dependent Types*. Kraus holds a University Research Fellowship, the Royal Society's 8-year flagship fellowship for early career researchers, and an ERC Consolidator grant (*Translations between Type Theories*). In total, the leadership team has supervised 7 postdoctoral researchers as well as 13 PhD students to completion.

We have a huge network of international collaborations all over the world, including our carefully selected project partners. Altenkirch holds a visiting position at Eötvös Loránd University. We are individually frequently invited to participate at major research meetings at centres including IAS Princeton, IHP Paris, Hausdorff Institute Bonn, Schloss Dagstuhl, CIRM Luminy, and CAS Oslo.

Module 3 – Contributions to the wider research and innovation community

We have given numerous invited or keynote talks (in 2025 at BCTCS and at the Logic Colloquium, as well as a few smaller workshops and seminars), invited lecture series (e.g. Days In Logic in Lisbon, 2024), served on program committees at leading conferences in theoretical computer science (in 2025, FSCD and LICS), and acted as opponents at many international PhD defences (in 2025: Stockholm, Budapest, Gothenburg, and Paris). We are particularly actively involved in the TYPES community, where Altenkirch (PI/lead) further served on the management committee for the COST action EUTYPES (CA15123), and is currently involved in the COST action EuroProofNet (CA20111). Altenkirch and Kraus participated in the special year on Univalent Foundations at the IAS Princeton (2012/13) and contributed to the HoTT Book. Altenkirch and Buchholtz frequently reach out to the philosophy community and have given invited talks at several philosophy workshops since 2021. Buchholtz is a contributor to the Lean proof assistant (<https://lean-lang.org/people>). We support the development of Agda by providing regular feedback, and teach Lean to our undergraduates.

Our group is a founding members of the Midlands Graduate School in the Foundations of Computing Science (MGS), a spring school for PhD students, early career researchers, and industry participants that has been running annually since 1999. In recent years, MGS attracts more than one hundred in-person participants. We both organise (most recently, 2018 and 2022)

and lecture (every year) at MGS. This allows us to reach interested people who are not yet members of our wider community, and places us in a perfect position to recruit PhD students and the PDRA requested as part of this application.

Module 4 – Contributions to broader research/innovation-users and audiences and towards wider societal benefit

Although we work in a deeply foundational field where direct societal impact may not be expected, we work hard on reaching out to non-experts and interested members of the public. One major medium for this is **Computerphile**, a popular YouTube channel with 2.3M subscribers that aims to explain computer science, programming, and related topics in an accessible way. Altenkirch has contributed 18 videos, one of which have breached 1M views and explain a selection of theoretical computer science and type-theoretic topics.

While the spring school MGS focusses on theory, it consistently attracts 10–20 industry participants per year and sponsorship from Meta, Huawei, Standard Chartered, and others. This is unsurprising given that type-theoretic methods are deployed in security-critical contexts. A well-known example is Google Chrome, where the encryption algorithm relies on code extracted from the Coq/Rocq proof assistant (Erbsen et al., 2019; cf. <https://news.mit.edu/2019/fiat-cryptography-chrome-android-0617>).

Agda plays a prominent role in industry applications: at Input Output (IOHK), it underpins the machine-checked formalisation of the Cardano ledger (Chapman et al., 2019) and the semantics of the Plutus and Marlowe smart contract languages. Type-theoretic verification ensures the correctness of consensus protocols, smart contract interpreters, and compilers, providing strong safety and liveness guarantees for distributed systems.

A further example of outreach is the HoTTeST summer school 2022, where Buchholtz lectured. The event attracted roughly 2,000 participants—many of them non-specialists—who engaged with homotopy type theory and joined the wider community.

6. References

Application question

List the references you have used to support your application.

Include all references in this section, not in the rest of the application questions.

You should not include any other information in this section.

We advise you not to include hyperlinks, as assessors are not obliged to access the information they lead to or consider it in their assessment of your application.

If linking to web resources, to maintain the information's integrity, include persistent identifiers (such as digital object identifiers) where possible.

You must not include links to web resources to extend your application.

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7. Project partners

Project partner	Contact	Direct contribution	Indirect contribution
New York University Abu Dhabi Abu Dhabi, United Arab Emirates	Urs Schreiber us13@nyu.edu		£5,000 Schreiber research time
Eötvös Loránd University Budapest, Hungary	Ambrus Kaposi akaposi@inf.elte.hu		£20,000 Kaposi research group
University of San Diego San Diego, United States	Michael Shulman shulman@sandiego.edu		£9,000 Shulmann research time
Johns Hopkins University Baltimore, United States	Emily Riehl erihl@jhu.edu		£6,300 Riehl research time
Reykjavík University Reykjavík, Iceland	Tarmo Uustalu tarmo@cs.ioc.ee		£30,800 Uustalu and Neumann research time
Utrecht University Utrecht, Netherlands	Paige North p.r.north@uu.nl		£5,000 North research time
Chapman University Orange, United States	Jonathan Weinberger jweinberger@chapman.edu		£5,300 Weinberger research time
Aarhus University Århus, Denmark	Daniel Gratzer danny.gratzer@gmail.com		£8,000 Gratzer research time

8. Project partners: letters (or emails) of support

Application question

Upload a single PDF containing the letters or emails of support from each partner you named in the previous Project Partners section.

If you do not have any project partners, simply add 'N/A' into the text box, mark this section as complete and move to the next section.

If you have named project partners in the previous Project Partners section, enter the words 'attachment supplied' in the text box.

Each letter or email you provide should:

- confirm the partner's commitment to the project
- clearly explain the value, relevance and possible benefits of the work to them
- describe any additional value that they bring to the project
- refer to EPSRC's guidance on project partners letters of support for further information [Project partners letter of support – UKRI](https://www.ukri.org/councils/epsrc/guidance-for-applicants/what-to-include-in-your-proposal/project-partners-letter-of-support/) (<https://www.ukri.org/councils/epsrc/guidance-for-applicants/what-to-include-in-your-proposal/project-partners-letter-of-support/>).

Save letters or emails of support from each project partner you have listed in your application in a single PDF no bigger than 8MB. Unless specially requested, please do not include any personal data within the attachment.

Ensure you have prior agreement from project partners so that, if you are offered funding, they will support your project as indicated in the contributions template.

Do not provide letters of support from host and project co-leads' research organisations.

see attached

[letters-merged.pdf](#)

1.27MB

9. Facilities

Application question

Does your proposed research require the support and use of a facility?

If you will need to use a facility, follow your proposed facility's normal access request procedures. Ensure you have prior agreement so that if you

are offered funding, they will support the use of their facility on your project.

For each requested facility you will need to provide the:

- name of facility, copied and pasted from the [facility information list \(DOCX, 42KB\)](https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/Facility+Information+for+TFS+updated+Sept+24.docx) (<https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/Facility+Information+for+TFS+updated+Sept+24.docx>).
- proposed usage or costs, or costs per unit where indicated on the facility information list
- confirmation you have their agreement where required

Write 'N/A' if you do not need to use a facility.

N/A

10. Sensitive information

Application question

Is there sensitive information you need to share with UKRI that you do not want shared with assessors?

If you do not have anything to share, enter 'No' into the text box, mark this section as complete and move on to the next section.

If you, or a key team member, need to tell us something you wish to remain confidential, enter 'Yes' in the text box.

Then email EPSRC on tfschangeepsr@epsr.ukri.org You must include in the subject line: <EPSRC responsive mode, sensitive info, Opportunity number, Application number>.

Typical examples of confidential information include:

- applicant is unavailable until a certain date (for example due to parental leave)
- declaration of interest
- additional information about eligibility to apply that would not be appropriately shared in the Applicant and team capability section

- conflict of interest for UKRI to consider in reviewer or panel participant selection
- the application is an invited resubmission

For information about how UKRI handles personal data, see [UKRI's privacy notice](https://www.ukri.org/about-us/privacy-notice/) (<https://www.ukri.org/about-us/privacy-notice/>).

No

11. Resources and Cost

Total full economic costs (fEC)

£1,002,447

Total contribution from applying organisation(s)

£200,489.40

Total funding applied for

£801,957.60

Directly allocated

Cost categories

Category	Applied for
Staff	£170,712.00
Estates	£55,001.60
Other	£1,512.80

Staff breakdown

Name	Role	%FTE
Thorsten Altenkirch	Project lead	20%
Nicolai Kraus	Project co-lead (UK)	20%

Name	Role	%FTE
Ulrik Buchholtz	Project co-lead (UK)	20%

Directly incurred

Cost categories

Category	Applied for
Staff	£204,999.20
Equipment	-
Travel and subsistence	£48,800.00
Other	-

Staff breakdown

Name	Role	%FTE
PDRA	Research and innovation associate	100%

Indirect

Cost categories

Category	Applied for
Indirect costs	£320,932.00

Staff breakdown

No staff for this category

Exceptions

Cost categories

Category	Applied for
Staff	-
Equipment	-
Travel and subsistence	-
Other	-

Staff breakdown

No staff for this category

[Full cost breakdown by organisation \(opens in a new tab\)](#)

(/applications/APP82661/resources-and-costs/detailed-view?backlinkHref=%2Fapplications%2FAPP82661%3Ftab%3DreadTab&backlinkTitle=Back+to+Application).

Justification of resources

Justification of Resources

This document provides a justification of the resources requested in this proposal. Costs are grouped under staff, travel and subsistence, workshops and open participation, and other directly incurred costs. Estates and indirect costs are determined by the host institutions and require no further justification.

Staff

The largest component of the requested resources is staff time. We request support for one full-time Research Associate (RA) for four years. The RA will carry out the day-to-day research programme, integrate the different strands of the project, and ensure continuity across its duration. They will contribute both to the development of new theoretical results and to their formalisation in proof assistants, which is central to the proposal. The RA ensures sustained and dedicated research time beyond that available to the investigators

and provides valuable training for an early-career researcher in a strategically important area.

In addition, the budget requests 20% FTE for each of the three Investigators (Thorsten, Ulrik, and Nicolai). The investigators will undertake the bulk of the intellectual research. The work requires expertise in both category theory and type theory, and progress relies on sustained collaboration between researchers with complementary strengths in these areas. In parallel, the investigators will supervise and mentor the RA, ensuring that the RA's contributions are well integrated into the overarching research programme. They will also devote time to coordinating with partners and disseminating results to the community. The combination of substantial PI/Co-I input and a dedicated RA ensures both the depth of investigator-led innovation and the continuity and capacity needed for a project of this scale.

Travel and Subsistence

We request travel and subsistence to enable collaboration, dissemination, and community engagement. These activities are essential to achieving scientific impact and keeping the project internationally connected.

We budget for two structured hybrid meetings with collaborators: an early planning meeting and a mid-project feedback meeting. These ensure that the project benefits from external expertise and remains aligned with both foundational and application-oriented goals. We also request travel support for five key collaborators (£10,000), enabling their direct contribution to shaping the project.

We further request funding for conference and workshop travel for the core team. Each of the four team members will attend one major international conference outside Europe (£3,000 est.) and three European conferences over the project's lifetime (£2,000 est. each), giving a total of £36,000. These trips target leading venues — POPL, LICS, TYPES, FSCD, CT, CSL, and HoTT/UF — which are the main fora for theoretical computer science, type theory, and category theory. Participation allows us to present results, obtain feedback from experts, and maintain research networks.

We also include participation in the Agda Implementers' Meetings (AIM), which are essential community workshops for advancing formalisation tools. We plan for four attendances over the lifetime of the project, budgeted at £1,500 (.est) each (total £6,000). AIM provides direct contact with developers and expert users of Agda, ensuring that our formalisation work both benefits from and contributes to the state of the art.

Workshops and Open Participation

We request resources to organise a dissemination workshop towards the end of the project. This event will bring together project members, collaborators, and the wider research community to consolidate outcomes and explore future directions. We budget £9,000 in total: this covers five external participants (speakers and early-career researchers, at £1,500 each), plus catering (£700) and a conference dinner (£800).

The workshop will provide a forum to present results to a broad audience, support sustained discussion across type theory, category theory, and formal verification, and encourage inclusivity. The budget includes support for early-career researchers who might not otherwise be able to attend, helping to strengthen the long-term vitality of the community.

Other Directly Incurred Costs

We request only minimal support in this category, namely hybrid participation facilities to allow collaborators to join meetings remotely. No equipment is requested, as the host institutions provide all necessary infrastructure.

Estates and Indirect Costs

These are calculated by the host institutions according to standard methodologies and require no further justification.

Alignment with Vision

The requested resources are closely aligned with the project's aims, with each item contributing directly to the research programme.

The investigators' time is central to the project. They will carry out the bulk of the intellectual research, drawing on complementary expertise in category theory and type theory. Their direct involvement is essential: the project requires sustained collaboration between senior researchers to develop definitions, prove results, and connect theory with formalisation. The RA provides continuity and additional capacity, but the originality of the work depends primarily on the investigators' contributions.

Travel resources support this investigator-led research by enabling exchange of ideas with collaborators and peers. Meetings, conference presentations, and participation in AIM ensure that the investigators' results are tested, refined, and communicated in the international research community, and that the formalisation aspects feed back into tool development. The dissemination workshop provides a cost-effective way to consolidate outputs and extend their influence. By engaging both established leaders and early-career researchers, it strengthens the community around the themes of the project.

Overall, the resources requested form a focused budget. They secure the investigators' active involvement while providing the support needed to sustain progress, collaboration, and dissemination. The project is designed to achieve its goals efficiently and inclusively, with long-term value to the research community.

12. Ethics and responsible research and innovation (RRI)

Application question

What are the ethical or RRI implications and issues relating to the proposed work? If you do not think that the proposed work raises any ethical or RRI issues, explain why.

Demonstrate that you have identified and evaluated:

- the relevant ethical or responsible research and innovation considerations

- how you will manage these considerations

Within this section you can also demonstrate elements of your responses in visual form if relevant. You should:

- use images sparingly and only to convey important information that cannot easily be put into words
- insert each new image onto a new line
- provide a descriptive legend for each image immediately underneath it (this counts towards your word limit)
- ensure files are smaller than 8MB and in JPEG, JPG, JPE, JFI, JIF, JFIF, PNG, GIF, BMP or WEBP format

The proposed research is foundational and theoretical in nature. Because of this, we do not envisage direct ethical issues. In particular, the research will not require human participation, personal data, animals, or any form of biological samples. Nevertheless, we agree that responsibility in research is crucial.

To promote good practice in research, we act in accordance with the guidelines of the University of Nottingham, e.g. by observing the research concordat to support the development of PDRAs, and by ensuring a positive working environment where we will pay attention to wellbeing and training opportunities.

We are strongly convinced of the importance of open access research, which is why we will continue to make all our publications and accompanying formal proofs (software code) publicly available. All our publications are open access (gold or green route). Our institutions have agreements with major publishers (e.g. Cambridge University Press) waiving open access fees. In addition, we make preprints of all our papers publicly available on arXiv. Our software, meanwhile, is publicly accessible on Git repositories (on github.com or bitbucket.org) and, when accompanying a publication, is made available as an open source deposit on the archival website zenodo.org.

13. Genetic and biological risk

Application question**Does your proposed research involve any genetic or biological risk?**

In respect of animals, plants or microbes, are you proposing to:

- use genetic modification as an experimental tool, like studying gene function in a genetically modified organism
- release genetically modified organisms
- ultimately develop commercial and industrial genetically modified outcomes

If yes, provide the name of any required approving body and state if approval is already in place. If it is not, provide an indicative timeframe for obtaining the required approval.

Identify the organism or organisms as a plant, animal or microbe and specify the species and which of the three categories the research relates to.

Identify the genetic and biological risks resulting from the proposed research, their implications, and any mitigation you plan on taking. Assessors will want to know you have considered the risks and their implications to justify that any identified risks do not outweigh any benefits of the proposed research.

Write 'N/A' if this does not apply to your proposed work.

N/A

14. Research involving the use of animals

Application question**Does your proposed research involve the use of vertebrate animals or other organisms covered by the Animals Scientific Procedures Act?**

If you are proposing research that requires using animals, download and complete the [Animals Scientific Procedures Act template \(DOCX, 74KB\)](#).

(<https://ukri-tfs-prod-assets.s3.eu-west-2.amazonaws.com/AnimalResearchQuestionTemplate-BBSRC.docx>), which contains all the questions relating to research using vertebrate animals or other Animals (Scientific Procedures) Act 1986 regulated organisms. Then, save it as a PDF.

Write 'N/A' if this does not apply to your proposed work.

N/A

15. Conducting research with animals overseas

Application question

Will any of the proposed animal research be conducted overseas?

If you are proposing to conduct overseas research, it must be conducted in accordance with welfare standards consistent with those in the UK, as in Responsibility in the use of animals in bioscience research, page 14 (<https://www.nc3rs.org.uk/3rs-resources/responsibility-use-animals-bioscience-research>).

Ensure all named applicants in the UK and overseas are aware of this requirement. Provide a statement to confirm that:

- all named applicants are aware of the requirements and have agreed to abide by them
- this overseas research will be conducted in accordance with welfare standards consistent with the principles of UK legislation
- the expectation set out in Responsibility in the use of animals in bioscience research will be applied and maintained
- appropriate national and institutional approvals are in place

Overseas studies proposing to use non-human primates, cats, dogs, equines or pigs will be assessed during NC3Rs review (<https://www.nc3rs.org.uk/who-we-are-and-what-we-do#services>) of research applications. Provide the required information by completing the template from the question 'Research involving the use of animals'.

For studies involving other species, select, download, and complete the relevant Word checklist or checklists from this list:

- [Additional questions on the use of rodents overseas](https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20rodents%20overseas%20%286%29_0.docx)
(https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20rodents%20overseas%20%286%29_0.docx).
- [Additional questions on the use of rabbits overseas](https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20rabbits%20overseas.docx)
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20rabbits%20overseas.docx>).
- [Additional questions on the use of sheep overseas](https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20sheep%20overseas.docx)
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20sheep%20overseas.docx>).
- [Additional questions on the use of goats overseas](https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20goats%20overseas.docx)
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20goats%20overseas.docx>).
- [Additional questions on the use of pigs overseas](https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20pigs%20overseas.docx)
(<https://nc3rs.org.uk/sites/default/files/documents/Additional%20questions%20on%20the%20use%20of%20pigs%20overseas.docx>).
- [Additional questions on the use of cattle overseas](https://nc3rs.org.uk/sites/default/files/2021-12/Additional%20questions%20on%20the%20use%20of%20cattle%20overseas.docx)
(<https://nc3rs.org.uk/sites/default/files/2021-12/Additional%20questions%20on%20the%20use%20of%20cattle%20overseas.docx>).
- [Additional questions on the use of *Xenopus laevis* and *Xenopus tropicalis* overseas](https://nc3rs.org.uk/sites/default/files/Additional%20questions%20on%20the%20use%20of%20Xenopus%20laevis%20%26%20%20Xenopus%20tropicalis%20overseas.docx)
(<https://nc3rs.org.uk/sites/default/files/Additional%20questions%20on%20the%20use%20of%20Xenopus%20laevis%20%26%20%20Xenopus%20tropicalis%20overseas.docx>).

Save as a PDF. If you use more than one checklist, save it as a single PDF.

Write 'N/A' if this does not apply to your proposed work.

N/A

16. Research involving human participation

Application question

Will the project involve the use of human subjects or their personal information?

If you are proposing research that requires the involvement of human subjects, provide the name of any required approving body and whether approval is already in place.

Justify the number and the diversity of the participants involved, as well as any procedures.

Provide details of any areas of substantial or moderate severity of impact.

Write 'N/A' if this does not apply to your proposed work.

N/A

17. Research involving human tissues or biological samples

Application question**Does your proposed research involve the use of human tissues, or biological samples?**

If you are proposing work that involves human tissues or biological samples, provide the name of any required approving body and whether approval is already in place.

Justify the use of human tissue or biological samples specifying the nature and quantity of the material to be used and its source.

Write 'N/A' if this does not apply to your proposed work.

N/A
